

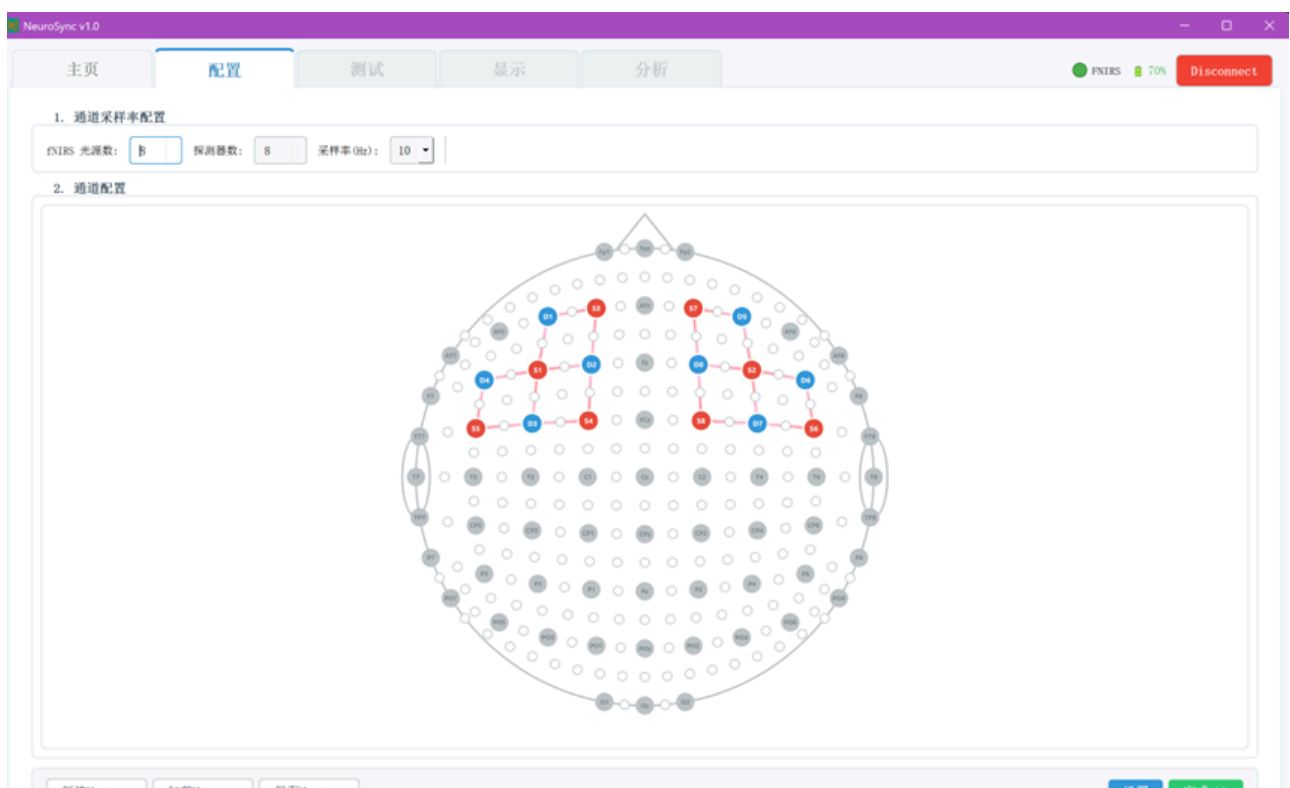
PyQt程序打包成exe

运行环境：

Windows 10 / VS Code / PowerShell

pyqt5、pyinstaller

要求：将下面软件转换为可执行的**exe**文件，发送给用户使用。



1.基础操作步骤

1.1.main.py为要转换的代码文件，icon中放着exe程序的图标



1.2.打开vscode，激活虚拟环境，进入正确的项目目录

```
PS D:\MyWorkspace\NeuroSync> &
d:\MyWorkspace\NeuroSync\.venv\Scripts\Activate.ps1
(.venv) PS D:\MyWorkspace\NeuroSync> cd NeuroSync
```

1.3.检查是否已安装PyInstaller

```
(.venv) PS D:\MyWorkspace\NeuroSync\NeuroSync> pip list | findstr
pyinstaller
```

终端显示:

```
WARNING: You are using pip version 22.0.4; however, version 26.0.1
is available.
You should consider upgrading via the
'd:\MyWorkspace\NeuroSync\.venv\Scripts\python.exe -m pip install -
--upgrade pip' command.
```

说明PyInstaller还没有安装。

1.4.先升级pip

```
(.venv) PS D:\MyWorkspace\NeuroSync\NeuroSync> python -m pip
install --upgrade pip
```

安装PyInstaller

```
(.venv) PS D:\MyWorkspace\NeuroSync\NeuroSync> pip install
pyinstaller
```

1.5.执行打包命令，希望包含常用隐藏导入

```
pyinstaller --onefile --windowed --name=NeuroSync --
icon=assets/icons/logo.ico --add-data="assets;assets" --hidden-
import=PyQt5.QtCore --hidden-import=PyQt5.QtGui --hidden-
import=PyQt5.QtWidgets --hidden-import=pyqtgraph --hidden-
import=pandas --hidden-import=numpy --hidden-import=scipy --hidden-
import=mne --hidden-import=pypinyin --hidden-import=h5py --hidden-
import=scipy.signal --hidden-import=logging.handlers main.py
```

2.报错的解决办法

2.1(缺少numpy)第一次打包后，运行生成的.exe文件，弹窗显示：

```
Traceback (most recent call last): File "main.py", line 8, in File "pyimod02_importers.py", line 457, in exec_module File "core\controller.py", line 10, in File "pyimod02_importers.py", line 457, in exec_module File "core\process.py", line 5, in File "pyimod02_importers.py", line 457, in exec_module File "utils\crc.py", line 1, in ModuleNotFoundError: No module named 'numpy'
```

说明打包在正常进行，但numpy模块缺失。

解决方案：

安装numpy模块(需要使用国内镜像加速)

```
pip install -i https://pypi.tuna.tsinghua.edu.cn/simple numpy
```

重新打包

```
pyinstaller build.spec
```

2.2(缺少pypinyin,等)第二次打包后，运行生成的.exe文件，弹窗显示：

```
Traceback (most recent call last): File "main.py", line 8, in File "pyimod02_importers.py", line 457, in exec_module File "core\controller.py", line 14, in File "pyimod02_importers.py", line 457, in exec_module File "core\widget_manager\user.py", line 10, in File "pyimod02_importers.py", line 457, in exec_module File "utils\subjects.py", line 7, in ModuleNotFoundError: No module named 'pypinyin'
```

说明打包在正常进行，但pypinyin模块缺失。

原因可能是PyInstaller在打包时无法自动检测到所有隐藏的依赖，特别是那些在运行时动态导入的模块。

运行这个命令，一次性安装所有需要的库：

```
pip install -i https://pypi.tuna.tsinghua.edu.cn/simple numpy pandas scipy h5py mne pyqtgraph psutil pypinyin
```

重新打包：

2.3(mne库的lazy_loader问题)第三次打包后，运行生成的.exe文件，弹窗显示：

Traceback (most recent call last): File "main.py", line 10, in File "", line 1007, in find_and_load File "", line 986, in find_and_load_unlocked File "", line 680, in load_unlocked File "pyimod02_importers.py", line 457, in exec_module File "mne_init.py", line 30, in File "lazy_loader_init.py", line 335, in attach_stub ValueError: Cannot load imports from non-existent stub 'C:\Users\25975\AppData\Local\Temp_MEI366962\mne\init.pyi'

说明已经解决了前面缺少库的问题。现在的问题是mne 库使用了 lazy_loader 机制，这导致 PyInstaller 打包时找不到 stub 文件。

①创建一个PyInstaller hook 文件来解决这个问题：

hook-mne.py

```
"""
PyInstaller hook for mne package
解决 mne 库的 lazy_loader 问题
"""

from PyInstaller.utils.hooks import collect_data_files,
collect_submodules

# 收集 mne 的所有子模块
hiddenimports = collect_submodules('mne')

# 收集 mne 的数据文件
datas = collect_data_files('mne', include_py_files=True)

# 排除 stub 文件相关模块，避免 lazy_loader 错误
excludedimports = ['mne._stubs', 'mne.stubs']

# 确保 mne 的关键模块被包含
hiddenimports += [
    'mne.io',
    'mne.io.array',
    'mne.io.edf',
    'mne.io.fif',
    'mne.preprocessing',
    'mne.filter',
    'mne.time_frequency',
```

```
    'mne.connectivity',  
    'mne.stats',  
    'mne.viz',  
]
```

hook-numpy.py

```
"""  
PyInstaller hook for numpy package  
解决 numpy 测试模块的依赖问题  
"""  
  
from PyInstaller.utils.hooks import collect_data_files,  
collect_submodules  
  
# 收集 numpy 的所有子模块  
hiddenimports = collect_submodules('numpy')  
  
# 收集 numpy 的数据文件  
datas = collect_data_files('numpy', include_py_files=True)  
  
# 确保 numpy.testing 模块被正确包含  
hiddenimports += [  
    'numpy.testing',  
    'numpy.testing.utils',  
    'numpy.testing.nosetester',  
    'numpy.testing.print_coercion_tables',  
]
```

hook-scipy.py

```
"""  
PyInstaller hook for scipy package  
解决 scipy 线性代数和信号处理模块的依赖问题  
"""  
  
from PyInstaller.utils.hooks import collect_data_files,  
collect_submodules  
  
# 收集 scipy 的所有子模块  
hiddenimports = collect_submodules('scipy')
```

```

# 收集 scipy 的数据文件
datas = collect_data_files('scipy', include_py_files=True)

# 确保 scipy 的关键模块被包含
hiddenimports += [
    'scipy.linalg',
    'scipy.linalg._basic',
    'scipy.linalg._decomp',
    'scipy.linalg._decomp_update',
    'scipy.linalg._matfuncs',
    'scipy.linalg._blas',
    'scipy.linalg._lapack',
    'scipy.signal',
    'scipy.signal.windows',
    'scipy.signal._spectral',
    'scipy.signal._filter_design',
    'scipy.signal._signaltools',
    'scipy.signal._waveforms',
    'scipy.signal._arraytools',
    'scipy.signal._upfirdn',
    'scipy.signal._peak_finding',
    'scipy.signal._savitzky_golay',
    'scipy.signal._spline',
    'scipy.signal._bsplines',
    'scipy.signal._filter_design',
    'scipy._lib',
    'scipy._lib._util',
    'scipy._lib._array_api',
    'scipy._lib.array_api_compat',
    'scipy._lib.array_api_compat.numpy',
]

```

②并更新build.spec文件，添加了hookspath指向，排除了有问题的stub模块

更新后的build.spec文件

```

# -*- mode: python ; coding: utf-8 -*-

block_cipher = None

```

```
a = Analysis(  
    ['main.py'],  
    pathex=[],  
    binaries=[],  
    datas=[  
        ('assets/styles.qss', 'assets'),  
        ('assets/icons/icon.png', 'assets/icons'),  
        ('assets/icons/logo.ico', 'assets/icons'),  
        ('assets/extinction_coefficients.csv', 'assets'),  
    ],  
    hiddenimports=[  
        # PyQt5相关  
        'PyQt5',  
        'PyQt5.QtWidgets',  
        'PyQt5.QtCore',  
        'PyQt5.QtGui',  
        'PyQt5.QtNetwork',  
        'PyQt5.QtWebEngineWidgets',  
  
        # 科学计算相关  
        'numpy',  
        'numpy.testing',  
        'numpy.testing.utils',  
        'pandas',  
        'scipy',  
        'scipy.linalg',  
        'scipy.signal',  
        'scipy.signal.windows',  
        'scipy._lib',  
        'scipy._lib._util',  
        'scipy._lib._array_api',  
        'h5py',  
        'mne',  
  
        # 其他第三方库  
        'pyqtgraph',  
        'pypinyin',  
        'psutil',  
        'uuid',  
        'json',  
    ]  
)
```

```
'csv',
'collections',
'enum',
'typing',
'dataclasses',
'logging',
'pathlib',
'datetime',
'time',
'socket',

# 自定义模块
'ui.mainWindow',
'core.controller',
'utils.logger',
'utils.paths',
'utils.crc',
'utils.filter',
'utils.stats',
'utils.subjects',
'physioSignal.fnirs',
'physioSignal.eeg',
'physioSignal.emg',
'core.thread_udp',
'core.buffer',
'core.process',
'core.widget_manager.bmap_manager',
'core.widget_manager.channel',
'core.widget_manager.display',
'core.widget_manager.quality',
'core.widget_manager.user',
'ui.views.channel_view',
'ui.views.display_view',
'ui.views.locate_widget',
'ui.views.quality_view',
'ui.views.user_view',
'ui.views.analysis_view',

# 确保 unittest 被包含, 因为 numpy/scipy 需要它
'unittest',
'unittest.mock',
'unittest.result',
```



```

        'unittest.case',
        'unittest.suite',
        'unittest.loader',
        'unittest.main',
        'unittest.runner',
    ],
    hookspath=['.'], # 使用当前目录的 hook 文件
    hooksconfig={},
    runtime_hooks=[],
    excludes=[
        # 排除 mne 的 stub 相关模块
        'mne._stubs',
        'mne.stubs',
        # 排除一些不需要的测试模块（但不包括 unittest，因为 numpy/scipy 需
要）
        'pytest',
        'test',
        'tests',
    ],
    win_no_prefer_redirects=False,
    win_private_assemblies=False,
    cipher=block_cipher,
    noarchive=False,
)

pyz = PYZ(a.pure, a.zipped_data, cipher=block_cipher)

exe = EXE(
    pyz,
    a.scripts,
    [],
    exclude_binaries=True,
    name='NeuroSync',
    debug=False,
    bootloader_ignore_signals=False,
    strip=False,
    upx=True,
    console=False,
    disable_windowed_traceback=False,
    argv_emulation=False,
    target_arch=None,
    codesign_identity=None,

```

```

        entitlements_file=None,
        icon='assets/icons/logo.ico',
    )

    coll = COLLECT(
        exe,
        a.binaries,
        a.zipfiles,
        a.datas,
        strip=False,
        upx=True,
        upx_exclude=[],
        name='NeuroSync',
    )

```

在main.py中插入以下显式导入：

```

# 显式导入所有可能用到的模块，帮助PyInstaller检测依赖
import numpy as np
import pandas as pd
import scipy
import scipy.signal
from scipy.signal import butter, sosfilt_zi, sosfilt, savgol_filter
import h5py
import mne # mne 的 lazy_loader 问题已通过 hook-mne.py 解决
import pyqtgraph
import pypinyin
import psutil
import uuid
import json
import csv
import collections
import enum
import typing
import dataclasses
import pathlib
import datetime
import time
import socket
import unittest

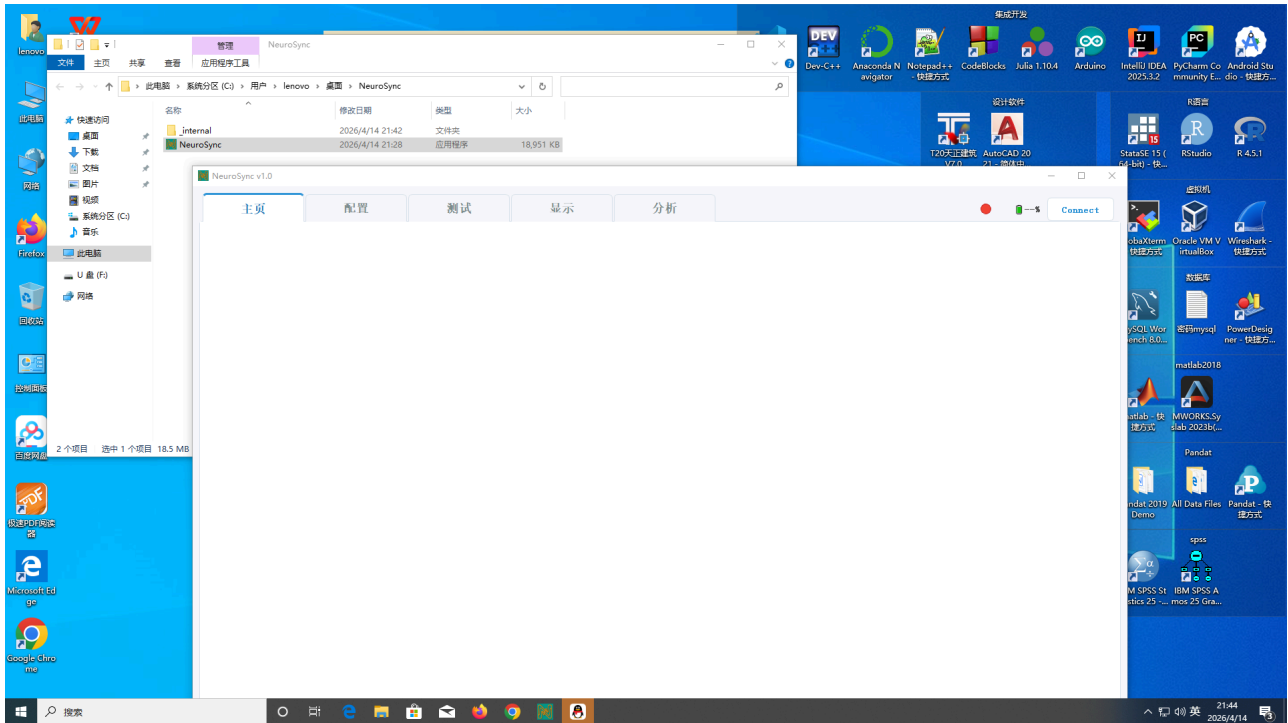
```

3.运行测试

最后运行没有报错：

```
pyinstaller build.spec
```

在不同环境(机房电脑)测试：



可以打开。